Designing a Model

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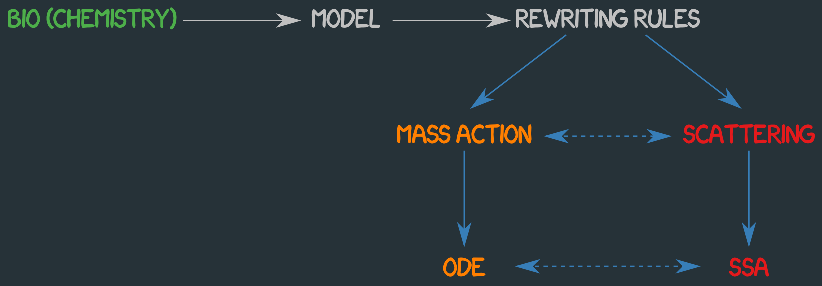
Design of mechanistic models



Rewriting rules

Chemical reactions are a language to describe transformations.

Take the reaction for the water formation:

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By looking at its compounds as if they were abstract objects:

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we describe a story of transformation:

when 2 elements A encounter 1 element C they are all transformed into 2 elements C

Rewriting rules

$$2\,A + B \longrightarrow 2\,C$$

The objects are the players and the reactions are their interactions.

The elements used to describe the story are:

- the alphabet $\{A, B, C\}$
- the set of rewriting rules (i.e. reactions) $2\,A + B \longrightarrow 2\,C$

Rewriting rules

$$\{A, B, X, Y, E\}$$

$$A \longrightarrow X$$
$$B + X \longrightarrow Y + D$$
$$2X + Y \longrightarrow 3X$$
$$X \longrightarrow E$$

- The rewriting rules are a “natural” and well spread representation of biochemical models
- they overlap with the stoichiometric system representation
- they are a unique and unambiguous representation for both deterministic and stochastic frameworks

From rewriting rules to matrices

$$A \longrightarrow X$$

$$B + X \longrightarrow Y + D$$

$$2X + Y \longrightarrow 3X$$

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From rewriting rules to matrices

$$
\begin{aligned}
A &\longrightarrow X \\
B + X &\longrightarrow Y + D \\
2X + Y &\longrightarrow 3X \\
X &\longrightarrow E
\end{aligned}
$$

$$
\begin{array}{ccc}
\alpha_{1,1}x_1 + \alpha_{1,2}x_2 + \cdots + \alpha_{1,N}x_N & \xrightarrow{c_1} & \beta_{1,1}x_1 + \beta_{1,2}x_2 + \cdots + \beta_{1,N}x_N \\
\alpha_{2,1}x_1 + \alpha_{2,2}x_2 + \cdots + \alpha_{2,N}x_N & \xrightarrow{c_2} & \beta_{2,1}x_1 + \beta_{2,2}x_2 + \cdots + \beta_{2,N}x_N \\
\vdots & \vdots & \vdots \\
\alpha_{M,1}x_1 + \alpha_{M,2}x_2 + \cdots + \alpha_{M,N}x_N & \xrightarrow{c_M} & \beta_{M,1}x_1 + \beta_{M,2}x_2 + \cdots + \beta_{M,N}x_N
\end{array}
$$

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\end{array}$$

From rewriting rules to matrices

$$
\begin{aligned}
\alpha_{1,1}x_1 + \alpha_{1,2}x_2 + \cdots + \alpha_{1,N}x_N &\xrightarrow{c_1} \beta_{1,1}x_1 + \beta_{1,2}x_2 + \cdots + \beta_{1,N}x_N \\
\alpha_{2,1}x_1 + \alpha_{2,2}x_2 + \cdots + \alpha_{2,N}x_N &\xrightarrow{c_2} \beta_{2,1}x_1 + \beta_{2,2}x_2 + \cdots + \beta_{2,N}x_N \\
\vdots \qquad\qquad & \vdots \qquad\qquad\qquad \vdots \\
\alpha_{M,1}x_1 + \alpha_{M,2}x_2 + \cdots + \alpha_{M,N}x_N &\xrightarrow{c_M} \beta_{M,1}x_1 + \beta_{M,2}x_2 + \cdots + \beta_{M,N}x_N
\end{aligned}
$$

$$
\begin{pmatrix}
\alpha_{1,1} & \alpha_{1,2} & \dots & \alpha_{1,N} \\
\alpha_{2,1} & \alpha_{2,2} & \dots & \alpha_{2,N} \\
\vdots & \vdots & \ddots & \vdots \\
\alpha_{M,1} & \alpha_{M,2} & \dots & \alpha_{M,N}
\end{pmatrix}
\qquad
\begin{pmatrix}
\beta_{1,1} & \beta_{1,2} & \dots & \beta_{1,N} \\
\beta_{2,1} & \beta_{2,2} & \dots & \beta_{2,N} \\
\vdots & \vdots & \ddots & \vdots \\
\beta_{M,1} & \beta_{M,2} & \dots & \beta_{M,N}
\end{pmatrix}
\qquad
\begin{pmatrix}
c_1 \\
c_2 \\
\vdots \\
c_M
\end{pmatrix}
$$

From rewriting rules to matrices

$$L = \begin{pmatrix} \alpha_{1,1} & \alpha_{1,2} & \cdots & \alpha_{1,N} \\ \alpha_{2,1} & \alpha_{2,2} & \cdots & \alpha_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{M,1} & \alpha_{M,2} & \cdots & \alpha_{M,N} \end{pmatrix}$$

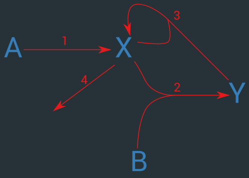
$$R = \begin{pmatrix} \beta_{1,1} & \beta_{1,2} & \cdots & \beta_{1,N} \\ \beta_{2,1} & \beta_{2,2} & \cdots & \beta_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{M,1} & \beta_{M,2} & \cdots & \beta_{M,N} \end{pmatrix}$$

$$\overrightarrow{c} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_M \end{pmatrix}$$

$$D = R - L = \begin{pmatrix} \delta_{1,1} & \delta_{1,2} & \cdots & \delta_{1,N} \\ \delta_{2,1} & \delta_{2,2} & \cdots & \delta_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ \delta_{M,1} & \delta_{M,2} & \cdots & \delta_{M,N} \end{pmatrix}$$

the BZ Reaction and the Brusselator

The Brusselator(JCP 46 1695) is a theoretical extremely simplified version (though physically unrealistic)
of the BZ (Belousov-Zhabotinskii) Reaction now days recognized as being the prototype for Chemical Oscillators:



$$A \longrightarrow X$$
$$B + X \longrightarrow Y + D$$
$$2X + Y \longrightarrow 3X$$
$$X \longrightarrow E$$

From rewriting rules to matrices

$$A \xrightarrow{c_1} X$$
$$B + X \xrightarrow{c_2} Y + D$$
$$2X + Y \xrightarrow{c_3} 3X$$
$$X \xrightarrow{c_4} E$$

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} \quad \vec{c} = \begin{pmatrix} 1 \\ 5\,10^{-3} \\ 2.5\,10^{-5} \\ 1.5 \end{pmatrix}$$

Not only bio-chemistry!

It is possible to exploit rewriting rules to describe any kind of transition.

Lotka-Volterra

The Lotka-Volterra model aims to describe predator prey interactions to simulate the populations dynamics.

In this model there are:

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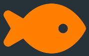
- preys 

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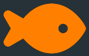
- preys 
- predators 

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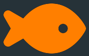
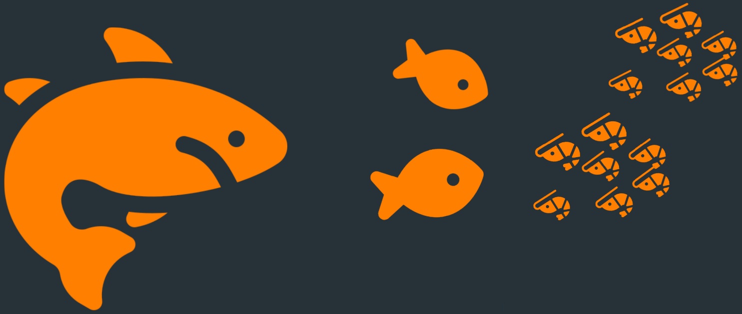
- preys 
- predators 
- prey’s food 

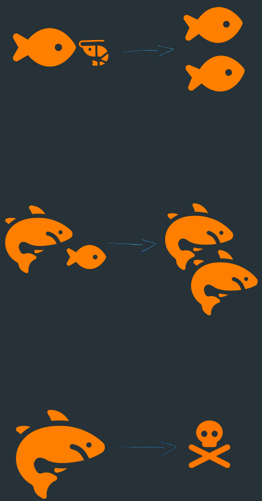
Lotka-Volterra

In a single patch:



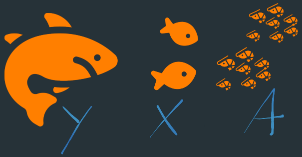
Lotka-Volterra

They interact:



Lotka-Volterra

By associating an alphabet it is possible to design a mathematical model by means of rewriting rules:



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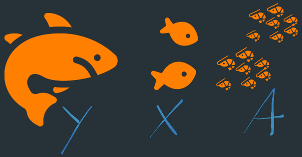
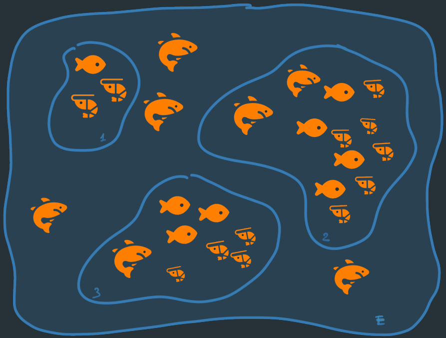


$$A + X \longrightarrow 2X$$
$$Y + X \longrightarrow 2Y$$
$$Y \longrightarrow$$

From Lotka-Volterra to metapopulations



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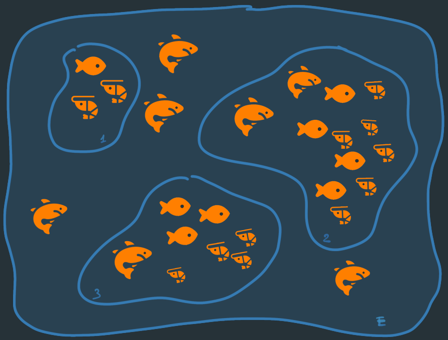

$$A_1 + X_1 \longrightarrow 2X_1$$
$$Y_1 + X_1 \longrightarrow 2Y_1$$
$$Y_1 \longrightarrow$$
$$A_2 + X_2 \longrightarrow 2X_2$$
$$Y_2 + X_2 \longrightarrow 2Y_2$$
$$Y_2 \longrightarrow$$
$$\dots$$
$$Y_1 \longrightarrow Y_E$$
$$Y_E \longrightarrow Y_1$$
$$Y_2 \longrightarrow Y_E$$
$$Y_E \longrightarrow Y_2$$
$$\dots$$